9. Compare Linear and Polynomial Regression using Python

a)

CGPA	Aptitude Score	Communication Score	Placement
9.2	90	88	Placed
8.8	85	82	Placed
8.5	80	79	Placed
7.2	72	70	Not Placed
6.8	68	65	Not Placed
9	92	90	Placed
7.5	75	74	Not Placed
8.6	84	81	Placed
6.5	60	58	Not Placed
8.9	88	86	Placed

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures, LabelEncoder
from sklearn.metrics import mean_squared_error, r2_score
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_1", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, [0]]
y = df.iloc[:, -1]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
linear_pred = linear_model.predict(X_test)
poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)
poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
poly_pred = poly_model.predict(X_test_poly)
print("\nLinear Regression")
```

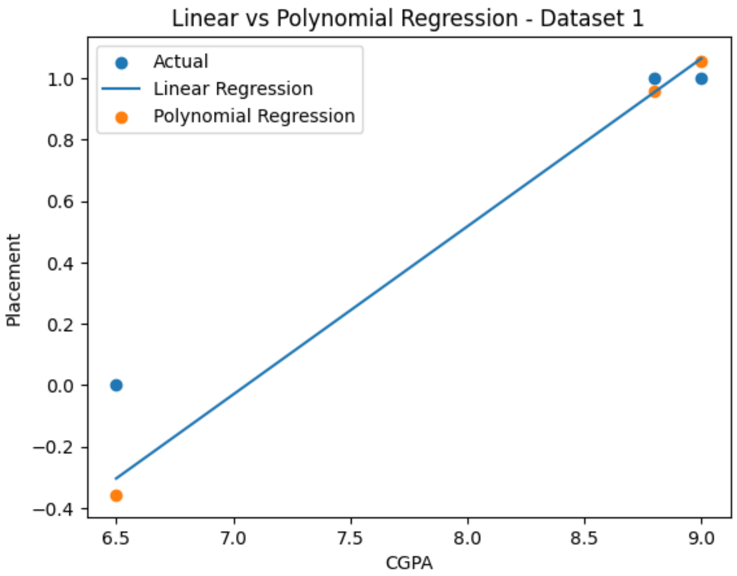
```
print("MSE:", mean_squared_error(y_test, linear_pred))
print("R2 Score:", r2_score(y_test, linear_pred))
print("\nPolynomial Regression")
print("MSE:", mean_squared_error(y_test, poly_pred))
print("R2 Score:", r2_score(y_test, poly_pred))
plt.scatter(X_test, y_test, label="Actual")
plt.plot(X_test, linear_pred, label="Linear Regression")
plt.scatter(X_test, poly_pred, label="Polynomial Regression")
plt.xlabel(df.columns[0])
plt.ylabel(df.columns[-1])
plt.title("Linear vs Polynomial Regression - Dataset 1")
plt.legend()
plt.show()
```

OUTPUT:

```
Dataset:
   CGPA  Aptitude Score  Communication Score   Placement
0   9.2              90                   88      Placed
1   8.8              85                   82      Placed
2   8.5              80                   79      Placed
3   7.2              72                   70  Not Placed
4   6.8              68                   65  Not Placed
5   9.0              92                   90      Placed
6   7.5              75                   74  Not Placed
7   8.6              84                   81      Placed
8   6.5              60                   58  Not Placed
9   8.9              88                   86      Placed

Linear Regression
MSE: 0.032765498777636105
R2 Score: 0.8525552555006375

Polynomial Regression
MSE: 0.044119842439841316
R2 Score: 0.8014607090207141
```



b)

Income (Lakhs)	Credit Score	Loan Amount (Lakhs)	Loan Approved
8	780	5	Yes
7	760	4	Yes
6	720	6	Yes
4	620	8	No
3	580	10	No
9	800	4	Yes
5	650	7	No
8	770	5	Yes
4	600	9	No
7	740	6	Yes

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures, LabelEncoder
from sklearn.metrics import mean_squared_error, r2_score
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_2", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, [0]]
y = df.iloc[:, -1]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
linear_pred = linear_model.predict(X_test)
poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)
poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
poly_pred = poly_model.predict(X_test_poly)
print("\nLinear Regression")
print("MSE:", mean_squared_error(y_test, linear_pred))
print("R2 Score:", r2_score(y_test, linear_pred))
print("\nPolynomial Regression")
print("MSE:", mean_squared_error(y_test, poly_pred))
```

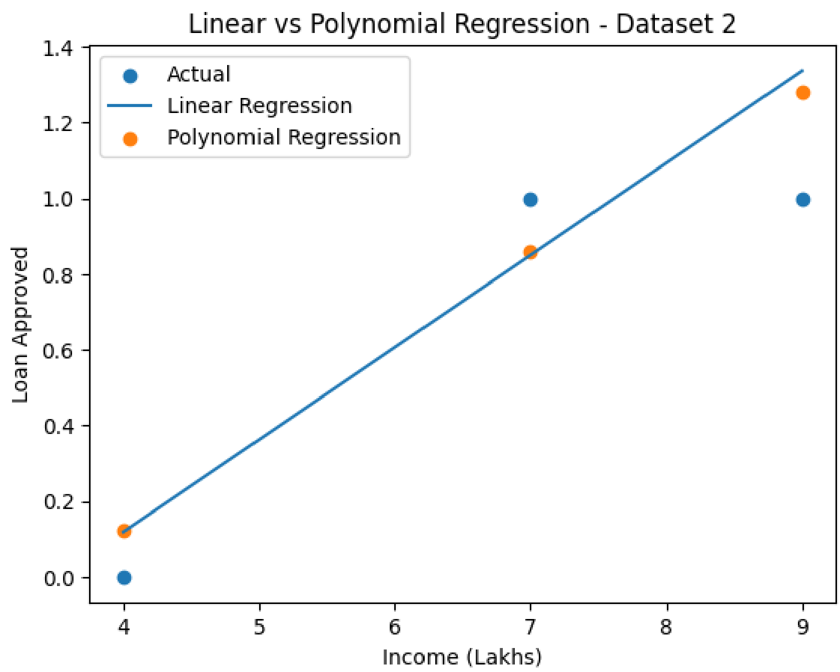
```
print("R2 Score:", r2_score(y_test, poly_pred))
plt.scatter(X_test, y_test, label="Actual")
plt.plot(X_test, linear_pred, label="Linear Regression")
plt.scatter(X_test, poly_pred, label="Polynomial Regression")
plt.xlabel(df.columns[0])
plt.ylabel(df.columns[-1])
plt.title("Linear vs Polynomial Regression - Dataset 2")
plt.legend()
plt.show()
```

OUTPUT:

```
Dataset:
   Income (Lakhs)  Credit Score  Loan Amount (Lakhs) Loan Approved
0               8           780                    5           Yes
1               7           760                    4           Yes
2               6           720                    6           Yes
3               4           620                    8            No
4               3           580                   10            No
5               9           800                    4           Yes
6               5           650                    7            No
7               8           770                    5           Yes
8               4           600                    9            No
9               7           740                    6           Yes

Linear Regression
MSE: 0.050169270833333245
R2 Score: 0.7742382812500004

Polynomial Regression
MSE: 0.03807829013958513
R2 Score: 0.8286476943718669
```



c)

Temperature (°C)	Heart Rate (bpm)	Oxygen Level (%)	Disease
39	110	94	Positive
38.5	108	95	Positive
37	78	99	Negative
39.2	115	93	Positive
36.8	75	98	Negative
38.8	112	94	Positive
37.2	80	98	Negative
39.1	114	92	Positive
36.7	74	99	Negative
38.9	111	93	Positive

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures, LabelEncoder
from sklearn.metrics import mean_squared_error, r2_score
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_3", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, [0]]
y = df.iloc[:, -1]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
linear_pred = linear_model.predict(X_test)
poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)
poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
poly_pred = poly_model.predict(X_test_poly)
print("\nLinear Regression")
print("MSE:", mean_squared_error(y_test, linear_pred))
print("R2 Score:", r2_score(y_test, linear_pred))
print("\nPolynomial Regression")
```

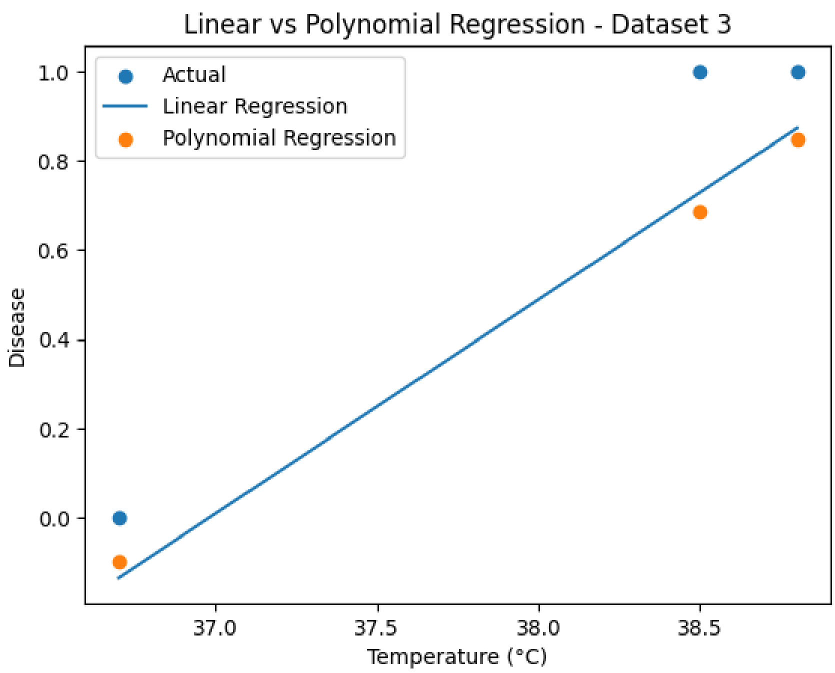
```
print("MSE:", mean_squared_error(y_test, poly_pred))
print("R2 Score:", r2_score(y_test, poly_pred))
plt.scatter(X_test, y_test, label="Actual")
plt.plot(X_test, linear_pred, label="Linear Regression")
plt.scatter(X_test, poly_pred, label="Polynomial Regression")
plt.xlabel(df.columns[0])
plt.ylabel(df.columns[-1])
plt.title("Linear vs Polynomial Regression - Dataset 3")
plt.legend()
plt.show()
```

OUTPUT:

```
Dataset:
   Temperature (°C)  Heart Rate (bpm)  Oxygen Level (%)   Disease
0              39.0               110                94  Positive
1              38.5               108                95  Positive
2              37.0                78                99  Negative
3              39.2               115                93  Positive
4              36.8                75                98  Negative
5              38.8               112                94  Positive
6              37.2                80                98  Negative
7              39.1               114                92  Positive
8              36.7                74                99  Negative
9              38.9               111                93  Positive

Linear Regression
MSE: 0.03586489534064465
R2 Score: 0.838607970967099

Polynomial Regression
MSE: 0.04412300737078626
R2 Score: 0.8014464668314618
```



d)

Experience	Performance Score	Training Hours	Promotion
10	95	50	Yes
8	90	45	Yes
7	88	40	Yes
3	65	20	No
2	60	18	No
9	93	48	Yes
4	70	25	No
8	89	42	Yes
2	58	15	No
6	85	38	Yes

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures, LabelEncoder
from sklearn.metrics import mean_squared_error, r2_score
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_4", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, [0]]
y = df.iloc[:, -1]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
linear_pred = linear_model.predict(X_test)
poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)
poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
poly_pred = poly_model.predict(X_test_poly)
print("\nLinear Regression")
print("MSE:", mean_squared_error(y_test, linear_pred))
print("R2 Score:", r2_score(y_test, linear_pred))
print("\nPolynomial Regression")
print("MSE:", mean_squared_error(y_test, poly_pred))
```

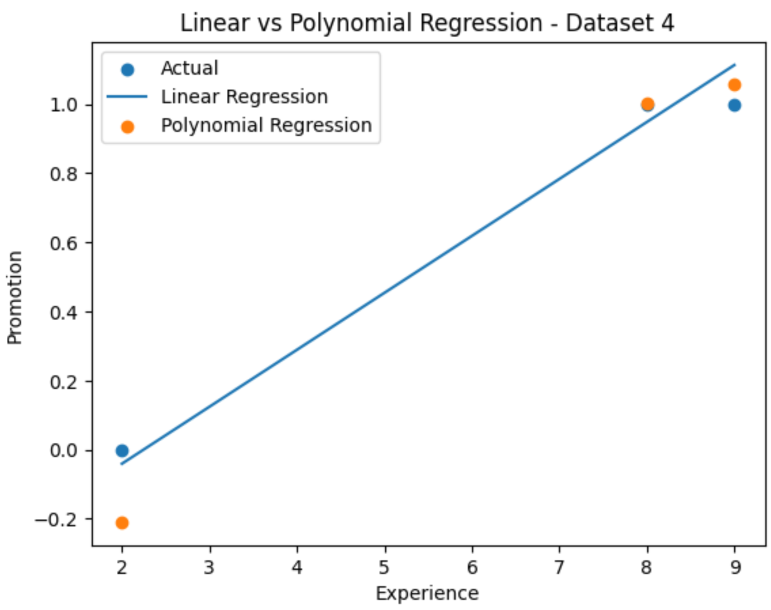
```
print("R2 Score:", r2_score(y_test, poly_pred))
plt.scatter(X_test, y_test, label="Actual")
plt.plot(X_test, linear_pred, label="Linear Regression")
plt.scatter(X_test, poly_pred, label="Polynomial Regression")
plt.xlabel(df.columns[0])
plt.ylabel(df.columns[-1])
plt.title("Linear vs Polynomial Regression - Dataset 4")
plt.legend()
plt.show()
```

OUTPUT:

```
Dataset:
   Experience  Performance Score  Training Hours Promotion
0          10                 95              50       Yes
1           8                 90              45       Yes
2           7                 88              40       Yes
3           3                 65              20        No
4           2                 60              18        No
5           9                 93              48       Yes
6           4                 70              25        No
7           8                 89              42       Yes
8           2                 58              15        No
9           6                 85              38       Yes

Linear Regression
MSE: 0.005682893960150141
R2 Score: 0.9744269771793244

Polynomial Regression
MSE: 0.015722933640396595
R2 Score: 0.9292467986182154
```



e)

Weight	Diameter	Sweetness	Fruit
150	7.5	90	Apple
160	7.8	88	Apple
140	7.2	91	Apple
120	5.5	70	Orange
125	5.8	72	Orange
130	6	74	Orange
155	7.6	89	Apple
118	5.4	69	Orange
148	7.4	92	Apple
122	5.7	71	Orange

PROGRAM:

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import PolynomialFeatures, LabelEncoder

from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_5", nrows=10)

print("Dataset:")

print(df)

X = df.iloc[:, [0]]

y = df.iloc[:, -1]

encoder = LabelEncoder()

y = encoder.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

linear_model = LinearRegression()

linear_model.fit(X_train, y_train)

linear_pred = linear_model.predict(X_test)

poly = PolynomialFeatures(degree=2)

X_train_poly = poly.fit_transform(X_train)

X_test_poly = poly.transform(X_test)
```

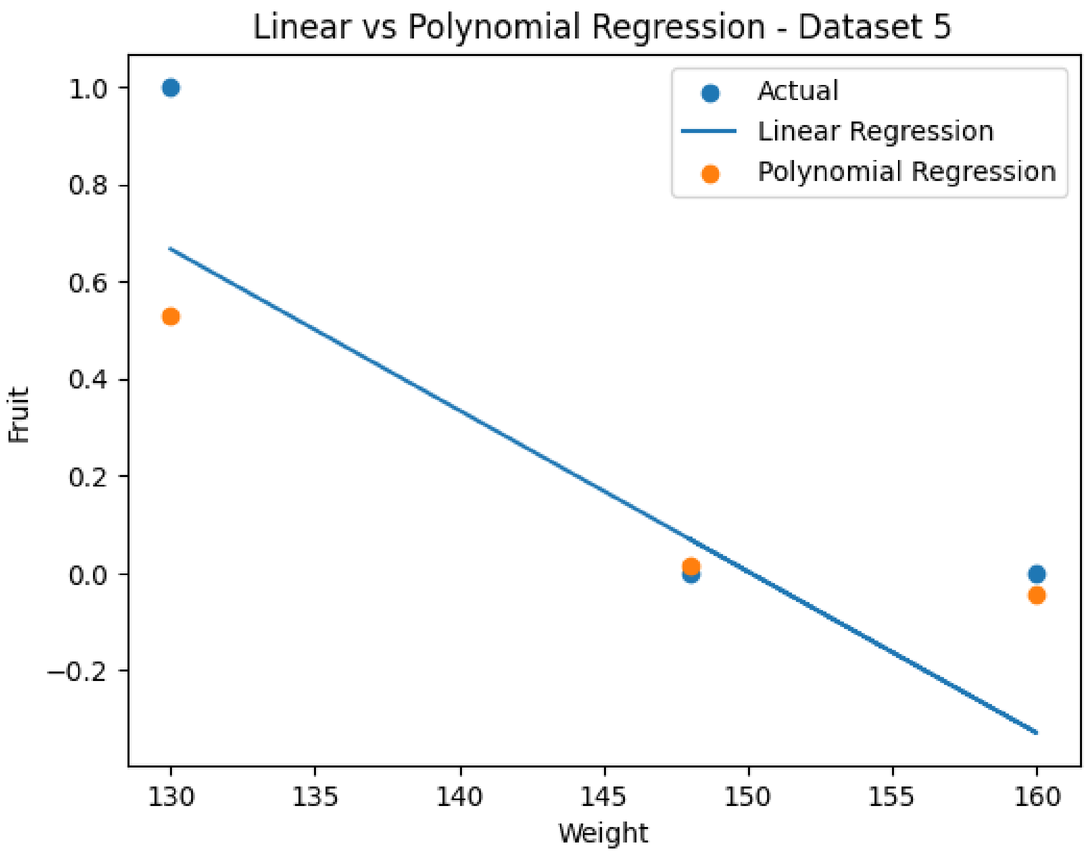
```
poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
poly_pred = poly_model.predict(X_test_poly)
print("\nLinear Regression")
print("MSE:", mean_squared_error(y_test, linear_pred))
print("R2 Score:", r2_score(y_test, linear_pred))
print("\nPolynomial Regression")
print("MSE:", mean_squared_error(y_test, poly_pred))
print("R2 Score:", r2_score(y_test, poly_pred))
plt.scatter(X_test, y_test, label="Actual")
plt.plot(X_test, linear_pred, label="Linear Regression")
plt.scatter(X_test, poly_pred, label="Polynomial Regression")
plt.xlabel(df.columns[0])
plt.ylabel(df.columns[-1])
plt.title("Linear vs Polynomial Regression - Dataset 5")
plt.legend()
plt.show()
```

OUTPUT:

```
Dataset:
   Weight  Diameter  Sweetness   Fruit
0     150       7.5         90   Apple
1     160       7.8         88   Apple
2     140       7.2         91   Apple
3     120       5.5         70  Orange
4     125       5.8         72  Orange
5     130       6.0         74  Orange
6     155       7.6         89   Apple
7     118       5.4         69  Orange
8     148       7.4         92   Apple
9     122       5.7         71  Orange

Linear Regression
MSE: 0.07466810020163138
R2 Score: 0.6639935490926588

Polynomial Regression
MSE: 0.07471854505772643
R2 Score: 0.6637665472402311
```


Linear vs Polynomial Regression - Dataset 5
Actual
Linear Regression
Polynomial Regression
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
Fruit
130
135
140
145
150
155
160
Weight